Multi-Tag UWB Localization With Spatial–Temporal Attention Graph Neural Network

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***Abstract*— Accurate ultrawideband (UWB) localization is still a challenging issue in complex environments. In this article, we propose a novel UWB localization method with twofold strategies to further improve its robustness and accuracy: 1) we construct the UWB ranging measurements into a graph structure and design a multiple tags UWB localization with the spatial–temporal attention graph neural network (STA-GNN-M) to handle it and 2) we utilize multiple tags to obtain multiple ranging measurements, comprehensively establish the geometric relationship between these measurements, and design a novel cost function to train the proposed network effectively. Our real-world experiments demonstrate that: 1) compared to the convolutional neural networks (CNNs) and long short-term memory (LSTM), which have been used in UWB localization, the graph neural network (GNN) can better capture the geometric relationships from the UWB ranging measurements, leading to a better estimation for tag location, and in addition, our proposed STA-GNN-M can effectively extract high-level spatial–temporal features from the measurements, which benefit UWB localization; and 2) using multiple tags can provide more information and geometric constraints, which can help improve the accuracy and robustness of UWB localization compare to the single tag-based methods, especially in non-line-of-sight (NLOS) environments.**

***Index Terms*— Graph neural network (GNN), multiple tags, spatial–temporal features, ultrawideband (UWB) localization.**

I. Introduction

ULTRAWIDEBAND (UWB) sensor is widely used for indoor localization due to its high time resolution and high accuracy [1], [2], [3], [4]. For accurate UWB indoor localization, time of arrival (TOA) [5], [6], [7] and time difference of arrival (TDOA) [8], [9], [10] are two commonly used methods since both of them can theoretically provide centimeter-level positioning accuracy. These two methods both handle the distance information: the TOA-based method directly uses the ranging measurements, while the TDOA-based method uses the difference of ranging measurements between difference anchors. Thus, at the algorithmic level, these two strategies are very similar for deep learning-based methods. In this article, we mainly focus on the TOA-based method due to the UWB sensors used in this work.

For TOA-based UWB localization, many efforts have been made in recent years. Some traditional methods, such as Kalman filter (KF)-based or optimization-based methods, have significantly improved its accuracy in line-of-sight (LOS) environments. Besides, deep learning-based methods [11], [12], [13], [14], such as convolutional neural networks (CNNs) and long short-term memory (LSTM), have also been introduced to improve the UWB localization performance in non-LOS (NLOS) environments. However, accurate UWB localization is still a challenging problem in a complex environment, where signals will be blocked by obstacles such as walls and furniture, resulting in large ranging errors. Based on these large errors, it is difficult for traditional methods to estimate accurate tag locations. In addition, although the existing deep learning-based methods achieve better performance than the traditional methods, they also suffer from some limitations as follows leading to undesirable localization results.

1) These methods only consider the ranging measurements and handle them as 2-D vectors. Besides, only the position values are used to design the cost function. However, the important geometric structures that provide the strict geometrical constraints among the anchor locations, ranging measurements, and tag locations are ignored.
2) When handling the ranging measurements from different anchors with different noises, their different impacts on localization are not considered sufficiently.
3) All of these methods use the distance information from only one tag, and however, it is not robust to locate in complex environments with too much interference based on only one tag.

In order to overcome these issues and further improve the UWB localization performance in complex environments, we propose a multiple tags UWB localization with the spatial–temporal attention graph neural network (STA-GNN-M). On the one hand, we first construct the UWB ranging measurements into a graph structure and then design a STA-GNN-M to handle it. Through the built graph, the topological structure of UWB sensors for localization can be considered. Besides, spatial and temporal features of distance information, which

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are both useful for UWB localization, can be extracted by our proposed STA-GNN-M. Meanwhile, the negative impacts of ranging measurements with large noise can also be suppressed through the attention layer in the proposed network. On the other hand, we utilize multiple tags in our localization, which can provide more ranging measurements for STA-GNN-M. Meanwhile, we design a multi-tag loss function containing distance constraints between the tags to further consider the geometric information in the training process. As a result, the accuracy and robustness of UWB localization are improved.

The main contributions of this article are summarized as follows.

1) A novel STA-GNN-M is proposed to improve the UWB localization performance in NLOS scenes. To the best of our knowledge, it is the first time that a graph neural network (GNN) has been used for UWB localization.
2) Multiple tags are first utilized for UWB localization to further improve its robustness and accuracy.
3) Real-world experiments are constructed to demonstrate the effectiveness of our proposed method.

The rest of this article is organized as follows. In Section II, we will briefly review the related work of our method. After that, the details of spatial–temporal graph construction, the architecture of designed STA-GNN-M, and the training strategy will be given in Section III. Then, we discuss the results of the real-world experiments in Section IV. Finally, the conclusion is given in Section V.

II. Related Works

A. Traditional TOA-Based UWB Localization

In terms of TOA-based UWB localization, the most basic approach is based on the least-squares method (LSM) [15], [16], [17]. The core of this method is to calculate a tag location that the distance between it and the fixed anchor positions has the smallest difference with the ranging measurements. Therefore, it is particularly susceptible to ranging errors and thus not suitable for complex environments. In order to overcome this issue, the optimization-based [3], [18], [19], [20] and KF-based methods [6], [21], [22] have been developed for UWB localization. As a representative work, a high-precision KF-based indoor 3-D UWB localization method is proposed in [23], where an improved particle swarm algorithm is designed to determine the parameters of the KF to improve the localization accuracy. Besides, some evolutionary algorithms are also introduced into the UWB localization. For example, an improved genetic algorithm is presented for UWB localization in [24]. In this work, a fitness function that includes intertag and anchor ranging information is constructed, while a Brownian motion-based search method is used to speed up the convergence. With these strategies, it achieves a good localization result in LOS scenes. However, the above traditional methods are unable to obtain a desirable performance in NLOS environments, where the ranging errors are larger and more random than those in LOS environments.

B. Deep Learning-Based UWB Localization

In recent years, deep learning-based UWB localization has been developed rapidly [11]. The core of these methods is to design a neural network to benefit the localization. Commonly, this designed network is trained with a large amount of collected data. In addition, root-mean-square error (RMSE) is generally used as the metric to evaluate the performance of different methods. As a presentative algorithm, the UWB localization system using a genetic algorithm and deep network is proposed in [25]. In this work, each anchor position is first effectively optimized by a genetic algorithm, and the LSTM model is mainly used to estimate ranging errors to obtain better ranging measurements. The supervised learning strategy with precollected ground-truth distance values are used to train this model. Finally, the tag locations are calculated based on the optimized anchor locations and ranging measurements. However, this method only considers the 2-D localization lacking the localization results in the z-axis, which is not suitable for situations with 3-D motion such as the localization for unmanned aerial vehicles. In addition, only using the LSTM model is hard to extract high-level features from the UWB measurements.

The state-of-the-art deep learning-based 3-D UWB localization method is the CNN-LSTM method proposed in [11]. This method first uses the CNN-based network to extract the spatial features from the ranging measurements in the same timestep and then feeds the extracted features from different timesteps to a deep LSTM model to further consider the correlations between consecutive timesteps. A supervised learning strategy based on mean square error cost function with position values is also designed to train the whole network. After the training process, it achieves a desirable performance in NLOS environments. However, this method as well as the existing end-to-end deep learning-based methods only consider the ranging measurements and handle them as 2-D vectors through the convolutional kernels or other neural networks. Besides, only the position values are used to design the cost function. The geometric structures, such as the relationships between the UWB anchors and tag, or the anchor locations, are ignored. The geometric structures are very important for the UWB localization since this information provides the strict geometrical constraints among the anchor locations, ranging measurements, and tag locations, and the UWB localization algorithm is developed based on these geometrical constraints. Thus, the lack of using the geometric information limits the performance of these UWB localization methods. In addition, all these methods only use one tag and do not consider the impact of different ranging noise on localization, which leads to limited feature extraction.

Therefore, our proposed method constructs the UWB ranging measurements into a graph structure to represent the geometric structures of UWB sensors. The spatial, temporal, and geometric relationships between the anchors and tags as well as the ranging measurements can be considered through the edges and nodes of the built graph; then, the whole graph can be handled through the proposed graph-based network, obtaining the high-level spatial–temporal features. In addition, the anchor locations are introduced into the network through the anchor position embedding process. Meanwhile, we use multiple tags in this work and design multiple loss functions to fully consider the important geometric information.

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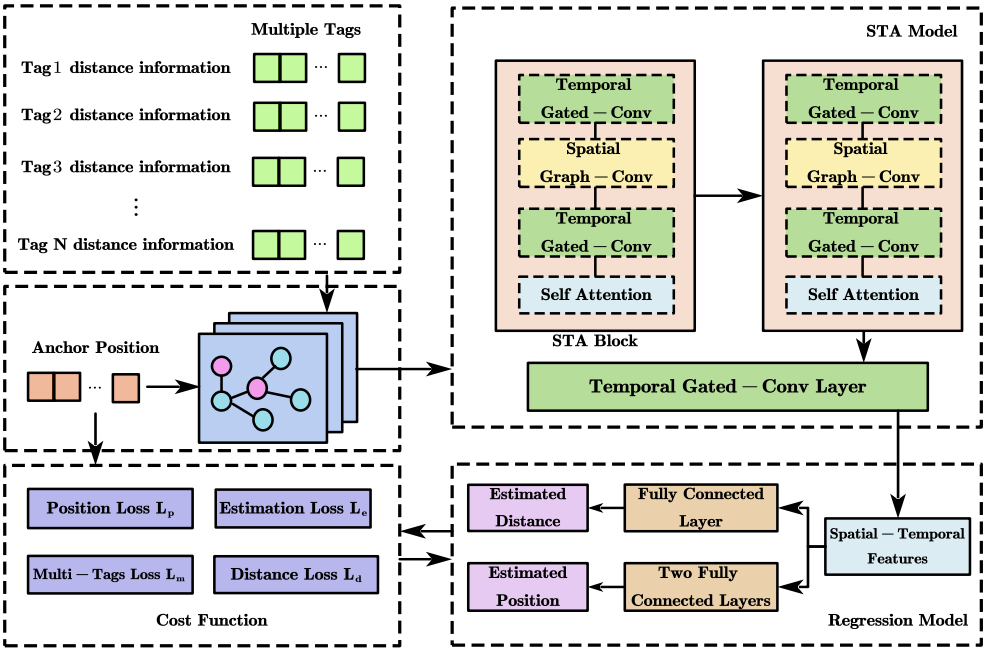


Fig. 1. Flowchart of our proposed STA-GNN-M: we mainly design the STA-GNN-M to handle the built spatial–temporal graph with collected distance information from multiple tags. Based on this proposed network, the tag location and the distance between the anchors and tags can be estimated simultaneously. In the training process, we design four losses to train the whole STA-GNN-M.

C. Graph Neural Network

GNN is proposed to handle the graph-structured data, which is mainly composed of nodes and edges [26], [27], [28]. The core of GNN is to enable nodes to learn high-dimensional features through neighbor aggregation and message passing mechanisms. Each node collects information from neighboring nodes and updates its own state through the training process [29]. Thus, it can extract high-level features from the graph-structured data.

Recently, graph networks have been used for localization. In [30], a floor localization method is proposed based on the GNN. A graph construction method is presented with the signal strength value of Wi-Fi, while a graph convolutional network is also introduced as a classifier to determine the floor of the target. For the UWB localization, the ranging measurements from the UWB sensors can be naturally represented as nodes of a graph, and the relationships between the measurements can be represented as edges. Thus, the GNN can be used to fuse and extract the high-level features from the ranging measurements. In this article, we design the STA-GNN-M to extract high-quality features and estimate the tag location.

III. Proposed Methods

Fig. 1 illustrates the flowchart of our proposed STA-GNN-M. First, based on the continuously ranging measurements provided by multiple tags, we construct a spatial–temporal graph. Besides, the embedding process is also built with fixed anchor positions. Second, the designed spatial–temporal attention (STA) model composed of temporal gated convolutional layer, spatial graph convolutional layer, and self-attention layer is utilized to extract high-level spatial–temporal features from the built graph. After that, the distance and tag locations are regressed through the fully connected layers. Finally, we design a novel cost function containing location loss, distance loss, estimation loss, and multi-tag loss to train the whole STA-GNN-M. We will introduce the graph construction process, STA-GNN-M, and the whole cost function in detail in Sections III-A–III-C.

A. Spatial–Temporal Graph Construction With Multiple Tags

In this article, we propose to use multiple tags to obtain more ranging measurements to help UWB localization. In order to provide more information to the STA-GNN-M, we collect ten consecutive ranging measurements between one tag and one anchor as a sample. If we install N tags on a carrier and M anchors in an environment, there will be $M \times N \times T$ samples during T timesteps, which can be denoted as $\boldsymbol{S} = \{\boldsymbol{s}_{1,1,1}, \boldsymbol{s}_{1,2,1}, \ldots, \boldsymbol{s}_{n,m,t}, \ldots, \boldsymbol{s}_{N,M,T}\}$, where m is the index of anchors, n is the index of tags, and t represents the timestep. When we estimate one tag location, all the related measurements from anchors and tags in the same timestep can be used to assist localization. Furthermore, the ranging measurements from different timesteps providing temporal information can also benefit the localization. Therefore, in order to fully use these related ranging measurements for UWB localization, we build a spatial–temporal graph as follows.

For each sample $\boldsymbol{s}_{n,m,t}$, we first make an anchor position embedding process for it. This process is similar to the position embedding process in Transformer [34]. To be specific, we first transform each sample and anchor into feature vectors with the same size through two independent fully connected layers and then add these two features as follows

$$\boldsymbol{fn}_{n,m,t} = \mathrm{FC}_s\left(\boldsymbol{s}_{n,m,t}\right) + \mathrm{FC}_a(\boldsymbol{A}_m) \tag{1}$$

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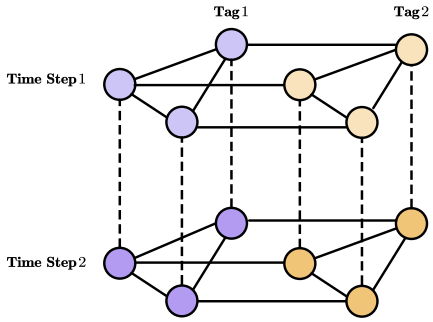


Fig. 2. Toy example of the spatial–temporal graph structure with three anchors and two tags: circles denote the node, solid lines denote the spatial edge, and dashed lines denote the temporal edge. It only shows two timesteps in this toy example.

where FC_s denotes the fully connected layer for samples, while FC_a denotes the fully connected layer for anchors. $\boldsymbol{A}_m$ is the position of the mth anchor. $\boldsymbol{fn}_{n,m,t} \in \mathbb{R}^D$ is the feature for each sample, where D is the feature dimension. These features can be viewed as a set of unordered nodes that are denoted as $\boldsymbol{Fnodes} = \{\boldsymbol{fn}_{1,1,1}, \boldsymbol{fn}_{1,2,1}, \ldots, \boldsymbol{fn}_{n,m,t}, \ldots, \boldsymbol{fn}_{N,M,T}\}$.

In terms of edges, we consider two types of edges in this work: spatial edge and temporal edge. For the nodes belonging to the same tag, we add an edge to connect them with each other. In addition, for the nodes from different tags, we only add an edge between the nodes containing the ranging measurements from the same anchor. The above edges are the spatial edge indicating the relationships between each ranging measurement in one timestep. For the temporal edge, it connects the nodes from the same anchor and tag between the adjacent timesteps. It can establish temporal relationships between ranging measurements. As a toy example, Fig. 2 shows the spatial–temporal graph structure with two timesteps based on three anchors and two tags, where circles denote the node, solid lines denote the spatial edge, and dashed lines denote the temporal edge.

The adjacency matrix $\boldsymbol{Adj} \in \mathbb{R}^{(M\times N\times T)\times(M\times N\times T)}$ of the built spatial–temporal graph can be represented as shown in Fig. 3(a), where $\mathbf{0}$ is the zero matrix and $\boldsymbol{I}$ is the identity matrix. Besides, $\boldsymbol{AS}_t \in \mathbb{R}^{(M\times N)\times(M\times N)}$ is the adjacency matrix of the spatial graph at timestep t shown in Fig. 3(b), where $\boldsymbol{O} \in \mathbb{R}^{N\times N}$ is a matrix with all elements being 1 except that the diagonal element is 0.

Through this spatial–temporal graph, we comprehensively consider both the spatial and temporal relationships between each ranging measurement for tag positioning. In addition, geometric information contained in the anchor positions is embedded into the input features through the anchor position embedding process.

B. Spatial–Temporal Attention Graph Neural Network

After constructing the spatial–temporal graph, we design the STA-GNN-M to handle it and estimate the tag location. This network consists of an STA model for feature extraction and a regression model for position and distance estimation. Besides, the STA model is mainly built by the temporal gated convolutional layer, spatial graph convolutional layer, and self-attention layer, while the regression model is based on fully

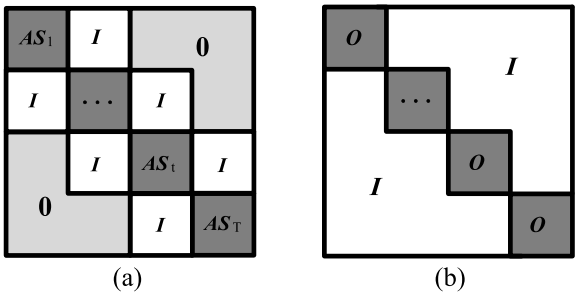


Fig. 3. (a) Adjacency matrix $\boldsymbol{Adj}$ of the spatial–temporal graph. (b) Adjacency matrix $\boldsymbol{AS}_t$ of the spatial graph at timestep t.

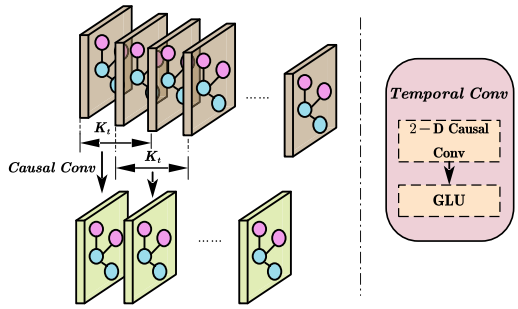


Fig. 4. Structure and flow of the temporal convolutional layer.

connected layers. We will detail these layers and models in the following.

1) Temporal Gated Convolutional Layer: Fig. 4 illustrates the structure and flowchart of the temporal gated convolutional layer. For this layer, only temporal edges are considered. Thus, for the input features $\boldsymbol{Fnodes}$, we first reshape it to $T \times D \times (M \times N)$ dimension. Next, we use a sliding window with length K_t and stride 1 to take the input data sequentially. For each taken data $\boldsymbol{F_r} \in \mathbb{R}^{K_t\times D\times(M\times N)}$, it can be seen as consisting of $(M \times N)$ subgraphs $\boldsymbol{F_{rg}} \in \mathbb{R}^{K_t\times D}$, whose nodes are connected by temporal edges. Then, a 2-D causal convolution operation [31] with C channels and $K_t \times D$ kernels are used to handle $\boldsymbol{F_{rg}}$ obtaining a C-dimension feature, as shown in Fig. 5. Thus, through the 2-D causal convolution operation, each node is fused with its temporal neighbors, obtaining a high-level fused feature $\boldsymbol{F_c} \in \mathbb{R}^{1\times C\times(M\times N)}$. After that, a gated linear unit (GLU) [32], which is adept at capturing long-term dependencies in the time series, is used as an activation function as follows:

$$\mathrm{GLU}(\boldsymbol{F_c}) = (\boldsymbol{W_1 F_c} + \boldsymbol{b_1}) \otimes \mathrm{sigmoid}(\boldsymbol{W_2 F_c} + \boldsymbol{b_2}) \quad (2)$$

where $\boldsymbol{W_1}, \boldsymbol{W_2} \in \mathbb{R}^{D'\times C}$ and $\boldsymbol{b_1}, \boldsymbol{b_2} \in \mathbb{R}^{D'}$ are learnable parameters, $\otimes$ denotes the element-wise product, and D' is the output feature dimension of GLU layer.

Through this temporal gated convolutional layer, each node and its temporal neighbors are fused, and thus, the temporal features can be learned. In addition, due to the sliding window operation, the size of the output feature is $(T - K_t + 1) \times D' \times (M \times N)$.

2) Spatial Graph Convolutional Layer: The spatial graph convolutional layer mainly considers the nodes from the same timestep. For a reshaped input data in the tth timestep

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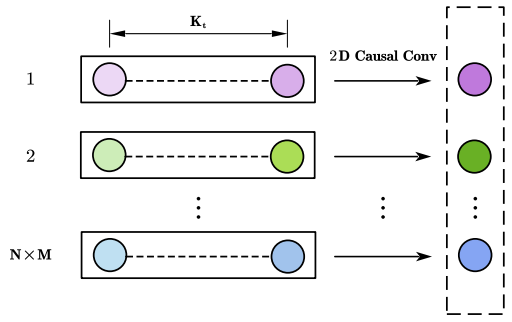


Fig. 5. Diagram of 2-D causal convolution operation.

$\boldsymbol{F}_t \in \mathbb{R}^{(M\times N)\times D'}$, this layer can be defined as follows:

$$\mathrm{SGC}(\boldsymbol{F}_t, \boldsymbol{AS}_t) = \sigma\left(\left(\boldsymbol{M} \otimes \tilde{\boldsymbol{D}}^{-\frac{1}{2}}\tilde{\boldsymbol{A}}\tilde{\boldsymbol{D}}^{-\frac{1}{2}}\right)\boldsymbol{F}_t\boldsymbol{W_3} + \boldsymbol{b_3}\right) \quad (3)$$

where $\boldsymbol{W_3} \in \mathbb{R}^{D\times D''}$ and $\boldsymbol{b_3} \in \mathbb{R}^{D''}$ are learnable parameters. Besides, $\tilde{\boldsymbol{A}} = \boldsymbol{AS}_t + \boldsymbol{I}$, $\tilde{\boldsymbol{D}}$ is the degree matrix of $\tilde{\boldsymbol{A}}$, $\boldsymbol{M} \in \mathbb{R}^{(M\times N)\times(M\times N)}$ is a learnable matrix, σ denotes the activation function, and D'' is the output feature dimension of each node.

This process is similar to the classical one-layer graph convolution neural network [33], but the learnable matrix $\boldsymbol{M}$ is introduced to learn the weight of each edge; since the UWB localization is a regression problem, learnable weights of each edge lead to a better regression result.

In addition, note that for the spatial graph convolutional layer in our STA model, if the input data contain several timesteps, this layer will handle the subgraph in each timestep.

Through this spatial graph convolutional layer, the features of each node in the same timestep are fused with its spatial neighbors, and thus, the spatial features from ranging measurements in the same timestep can be further extracted.

3) Self-Attention Layer: In terms of the self-attention layer, input features $\boldsymbol{F}$, including all the nodes, are first mapped to the query, key, and value spaces by different fully connected layers obtaining $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$. Then, the process of self-attention mechanism is defined as [34]

$$\mathrm{Att}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d_k}}\right)\boldsymbol{V} \quad (4)$$

where d_k is the feature dimension.

Finally, the output of the self-attention layer is defined as

$$\boldsymbol{F}_{\mathrm{att}} = \mathrm{LN}(\boldsymbol{F} + \mathrm{Att}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V})) \quad (5)$$

where LN denotes the layer normalization. These residual connections and layer normalization processes are used to address the vanishing gradient issue.

Self-attention mechanism is commonly used to assign attention weights to each feature and thus allows the model to focus on the features most important for the estimation. For our UWB localization, we feed all nodes of the graph into this layer, and thus, it can globally calculate the weight for each node. The nodes with large measurement noise will be assigned a small attention weight, and therefore, the negative effect of the nodes for the UWB localization can be suppressed.

4) STA Block: According to the constructed graph, the temporal relationship is more difficult to learn than the spatial relationship. Thus, as shown in Fig. 1, we use two temporal gated convolutional layers for the STA block. The spatial graph convolutional layer is used between two temporal gated convolutional layers, while the self-attention layer is in the end.

5) STA Model: The STA model is built with two STA blocks and one temporal gated convolutional layer. Through two STA blocks, each ranging measurement can be fused, large measurement noise can be suppressed, and high-level spatial–temporal features can be extracted. Besides, after two STA blocks, the size of input features $\boldsymbol{Fnodes}$ will become $(T - 4K_t + 4) \times D' \times (M \times N)$, due to the four temporal gated convolutional layers. Then, we use an additional temporal gated convolutional layer that the length of sliding window is $(T-4K_t+4)$ to fuse the nodes on all time series, obtaining the final spatial–temporal features $\boldsymbol{F}_{\mathrm{sta}} \in \mathbb{R}^{D'\times(M\times N)}$.

6) Regression Model: Finally, the regression model mainly constructed by the fully connected layers is used to estimate the distance information and tag location based on $\boldsymbol{F}_{\mathrm{sta}}$. To be specific, for estimating the distance information, we use one fully connected layer as

$$\boldsymbol{d}_{\mathrm{est},T} = \boldsymbol{W_4}\boldsymbol{F}_{\mathrm{sta}} + \boldsymbol{b_4} \quad (6)$$

where $\boldsymbol{W_4} \in \mathbb{R}^{1\times D'}$ and $\boldsymbol{b_4}$ are learnable parameters, and $\boldsymbol{d}_{\mathrm{est}} \in \mathbb{R}^{(M\times N)}$ is the estimated distance between N tags and M anchors at the Tth timestep.

In terms of the tag location, our aim is to estimate all N locations of tags. Hence, we first reshape the spatial–temporal features as $\boldsymbol{F}'_{\mathrm{sta}} \in \mathbb{R}^{(M\times D)\times N}$. Then, two fully connected layers are used as follows to calculate N tag locations:

$$\boldsymbol{p}_{\mathrm{est},T} = \left(\boldsymbol{W}_6\sigma\left(\boldsymbol{W_5}\boldsymbol{F}'_{\mathrm{sta}} + \boldsymbol{b}_5\right) + \boldsymbol{b}_6\right) \quad (7)$$

where $\boldsymbol{W_5} \in \mathbb{R}^{D'''\times(M\times D)}$, $\boldsymbol{W_6} \in \mathbb{R}^{3\times D'''}$, $\boldsymbol{b}_5$ and $\boldsymbol{b}_6$ are learnable parameters, and $\boldsymbol{p}_{\mathrm{est}} \in \mathbb{R}^{3\times N}$ is the estimated N tag locations at Tth timestep.

7) Spatial–Temporal Attention Graph Neural Network: The above processes constitute the whole flowchart of the STA-GNN-M. In practice, we first collect the ranging measurements from timestep 1 to T and build the spatial–temporal graph based on them. Then, we used the STA-GNN-M to estimate $\boldsymbol{d}_{\mathrm{est},T}$ and $\boldsymbol{p}_{\mathrm{est},T}$. Next, when we collect the ranging measurements in timestep $(T + 1)$, we utilize the data from timestep 2 to $(T + 1)$ to estimate $\boldsymbol{d}_{\mathrm{est},T+1}$ and $\boldsymbol{p}_{\mathrm{est},T+1}$. This process will continue until the localization is completed.

In addition, at a very low probability, a few anchors will lose their signals for a short period of time, due to the interference from multiple walls or furniture. In this situation, we use the ranging measurements in the last timestep for the input and do not change the length and order of the input signals.

C. Loss Function

In this work, we train the whole network based on the supervised learning strategy. The designed cost function contains four losses as follows.

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1) Position Loss: The position loss $\boldsymbol{L}_p$ builds the constraints between all estimated tag locations and their ground truth, which is defined as [11]

$$\boldsymbol{L}_p = \sum_{i=1}^{N} \left(\left\| \boldsymbol{p}_{\text{est},i} - \boldsymbol{p}_{\text{gt},i} \right\|_2 \right)^2 \tag{8}$$

where $\boldsymbol{p}_{\text{est},i} \in \mathbb{R}^3$ is the estimated ith tag location and $\boldsymbol{p}_{\text{gt},i}$ is its corresponding ground-truth value.

2) Distance Loss: Similar to the position loss, the distance loss $\boldsymbol{L}_d$ reflects the constraints between all reconstructed distances and their ground truth, which is defined as

$$\boldsymbol{L}_d = \sum_{j=1}^{M} \sum_{i=1}^{N} \left(d_{\text{est},ij} - d_{\text{gt},ij} \right)^2 \tag{9}$$

where $d_{\text{est},ij}$ is the reconstructed distance between the ith tag and the jth anchor, while $d_{\text{gt},ij}$ is its corresponding ground-truth value.

3) Multi-Tag Loss: The multiple tags utilized in this article are installed in fixed positions, and thus, the distances between these tags are also fixed. Therefore, we can build the multi-tag loss $\boldsymbol{L}_m$ based on these fixed distances and the calculated distances between the estimated tag locations

$$\boldsymbol{L}_m = \sum_{i=1}^{N} \sum_{k=i+1}^{N} \left(\left\| \boldsymbol{p}_{\text{est},i} - \boldsymbol{p}_{\text{est},k} \right\|_2 - d_{\text{tag},ik} \right)^2 \tag{10}$$

where $\boldsymbol{p}_{\text{est},i}$ and $\boldsymbol{p}_{\text{est},k}$ are the estimated ith and kth tag locations, respectively, and $\|\boldsymbol{p}_{\text{est},i} - \boldsymbol{p}_{\text{est},k}\|_2$ calculates the distance between these two locations. $d_{\text{tag},ik}$ is the true and fixed distance value between the ith and kth tag locations, which can be precalibrated.

4) Estimation Loss: In addition to the multi-tag loss, we build the estimation loss $\boldsymbol{L}_e$ to further consider the geometric information. The estimation loss is used to build the geometric constraints between the estimated distance and tag locations. The bridge of these two estimated values is the fixed anchor positions. Thus, the estimation loss is defined as follows:

$$\boldsymbol{L}_e = \sum_{j=1}^{M} \sum_{i=1}^{N} \left(d_{\text{est},ij} - \left\| \boldsymbol{p}_{\text{est},i} - \boldsymbol{A}_j \right\|_2 \right)^2 \tag{11}$$

where $\boldsymbol{A}_j$ is the position of the jth anchor. $\|\boldsymbol{p}_{\text{est},i} - \boldsymbol{A}_j\|_2$ calculates the distance between the ith tag and the jth anchor, while $d_{\text{est},ij}$ is the estimated distance between the ith tag and the jth anchor.

Based on the above losses, the overall cost function is defined as

$$\boldsymbol{L} = \alpha_1 \boldsymbol{L}_p + \alpha_2 \boldsymbol{L}_d + \alpha_3 \boldsymbol{L}_m + \alpha_4 \boldsymbol{L}_e \tag{12}$$

where α_1, α_2, α_3, and α_4 are the weights of losses.

These four loss functions are designed from different aspects, and they can provide different useful and consistent constraints to train the whole proposed network. Through this cost function, the output of the network can approach the ground-truth values, while the relative geometric information is introduced into the network. Thus, the high-level features can be extracted, while the accurate distances and tag locations can be regressed. In Section IV, we demonstrate that each loss included in the cost function benefits the UWB localization.

TABLE I
SITUATION OF EACH SEQUENCE IN THE EXPERIMENTS

Type of Scene	Sequence	Type	Duration	Number of Samples	Average measurement error
LOS	L-0	Train	845s	8404	0.18m
	L-1	Test	342s	3409	0.16m
	L-2	Test	346s	3455	0.17m
	L-3	Test	53s	522	0.14m
	L-4	Test	767s	7671	0.18m
	L-5	Test	338s	3388	0.16m
	L-6	Test	156s	1566	0.16m
	L-7	Test	377s	3752	0.18m
	L-8	Test	356s	3510	0.18m
	L-9	Test	305s	2977	0.17m
NLOS	N-0	Train	988s	9875	0.56m
	N-1	Test	698s	6981	0.59m
	N-2	Test	480s	4808	0.56m
	N-3	Test	54s	538	0.60m
	N-4	Test	57s	563	0.56m
	N-5	Test	97s	971	0.60m
	N-6	Test	186s	1859	0.54m
	N-7	Test	355s	3559	0.80m
	N-8	Test	446s	4430	0.79m
	N-9	Test	322s	3211	0.45m
	N-10	Test	384s	3809	0.45m
	N-11	Test	251s	2504	0.68m

IV. EVALUATION

A. Experimental Settings

In this section, we evaluate the proposed multi-tag UWB localization method with two constructed real-world experiments, since to the best of our knowledge, there are no public datasets involving multi-tag UWB data.

In our experiments, we use eight LinkTrack-P UWB sensors, and four of them are installed in the experimental environment as anchors ($N = 4$), while the other four sensors are installed on a mobile robot as tags ($M = 4$). Besides, an RS-Helios-5515 32-line LiDAR is also installed on the mobile robot. The 3-D LiDAR is used to provide the ground-truth trajectories with the LOAM method [35]. This LiDAR-based method can obtain centimeter-level localization results, which has errors of one order of magnitude smaller than the UWB localization. As shown in Fig. 6, the rotation and translation between the LiDAR coordinate system, tag coordinate systems and UWB coordinate system are precalibrated. Thus, when we obtain the trajectories of LiDAR sensors, we can generate the ground-truth trajectories of the four tags. Then, the ground-truth distances can be calculated with the ground-truth trajectories and the precalibrated anchor positions. In addition, the NVIDIA Jetson Orin NANO is used in the robot to control the sensors and execute the proposed localization method.

In terms of experimental environments, we build two experiments with different difficulty levels and anchor positions. Fig. 7 gives the structure map of the two experiments, while Table I gives the details of all sequences in our experiments. The average measurement error in Table I means the average differences between all the ranging measurements from UWB sensors and the ground-truth distances. It shows that the first

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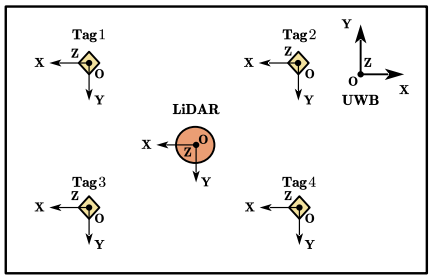


Fig. 6. Diagram of LiDAR coordinate system, tag coordinate systems, and UWB coordinate system.

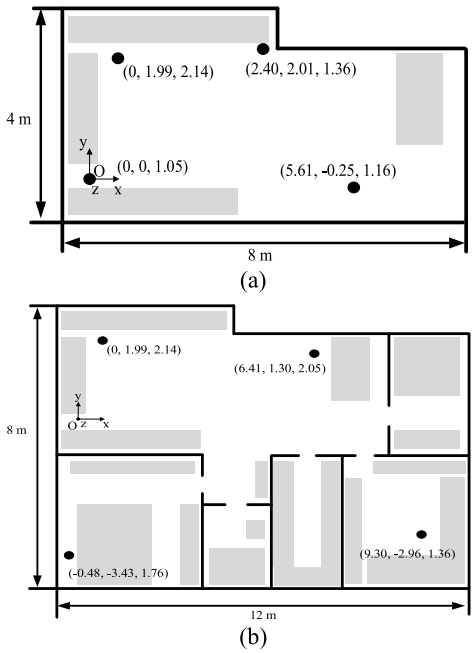


Fig. 7. Structure map of experiments and the corresponding anchor positions. (a) First experiment in living room and (b) second experiment in all residential sense. The gray rectangle denotes the furniture.

experiment is conducted in the living room, which is an LOS scene. Although the multipath effect will affect the ranging accuracy, its average ranging errors are still relatively small. The second experiment is built with all the residential scenes, where the signals of UWB sensors are influenced by a few walls and furniture. Therefore, it is an NLOS scene that the average measurement errors are large.

In each experiment, we control our mobile robot to move randomly in the environment. For the training sequences, we record the ranging measurements as well as the measurements from the 3-D LiDAR, while for the testing sequences, we also record the real-time 3-D localization results.

In terms of competing methods, the GA-based method [3] is used as a representative of traditional approaches. For the deep learning-based approaches, we choose the CNN-LSTM method [11] since it achieves state-of-the-art results. In addition, since the CNN-LSTM method only uses a single tag and $\boldsymbol{L}_p$ loss for training, we also build the STA-GNN-M with a single tag (STA-GNN-S) and $\boldsymbol{L}_p$ loss. For this method, we only feed the ranging measurements from the first tag to the network and only use the $\boldsymbol{L}_p$ loss as the cost function to train the network.

For the evaluation metric, since some competing methods only use one tag in their systems, we use the RMSE of the first tag (Tag 1) to evaluate the accuracy of our proposed multi-tag UWB localization method and competing methods.

Finally, the whole method is implemented by PyTorch and trained on the NVIDIA RTX 3060. The detailed parameters of different layers of our proposed network are shown in Table II. Besides, the learning rate is set as 0.001, while the batch size is set as 1024. In addition, we set $\alpha_1, \alpha_2, \alpha_3$, and α_4 as 1, 0.2, 0.2, and 0.2, respectively. We discuss this setting in Section IV-C. The training time is less than 30 min for both experiments. In terms of the input data, we consider 12 timesteps for one input ($T = 12$). The collection frequency of the UWB sensors is 100 Hz, and thus, the localization frequency is 10 Hz, which is the same as the ground truth. After the training process, the trained network is executed on the NVIDIA Jetson Orin NANO. In this device, the localization frequency of our proposed method can reach 10 Hz.

B. Results

The results of our proposed STA-GNN-M method and all the competing methods are given in Table III. In addition, in order to compare our proposed method with other method using the same cost function, the results of STA-GNN-M with only $\boldsymbol{L}_p$ loss are also shown in this table. Besides, Fig. 8 gives the trajectory errors of our proposed method and competing methods of L-7 sequence in the first experiment and N-9 sequence in the second experiment as examples, while Fig. 9 shows the cumulative distribution functions (CDFs) of localization errors of whole sequences in experiments indicating the overall accuracy.

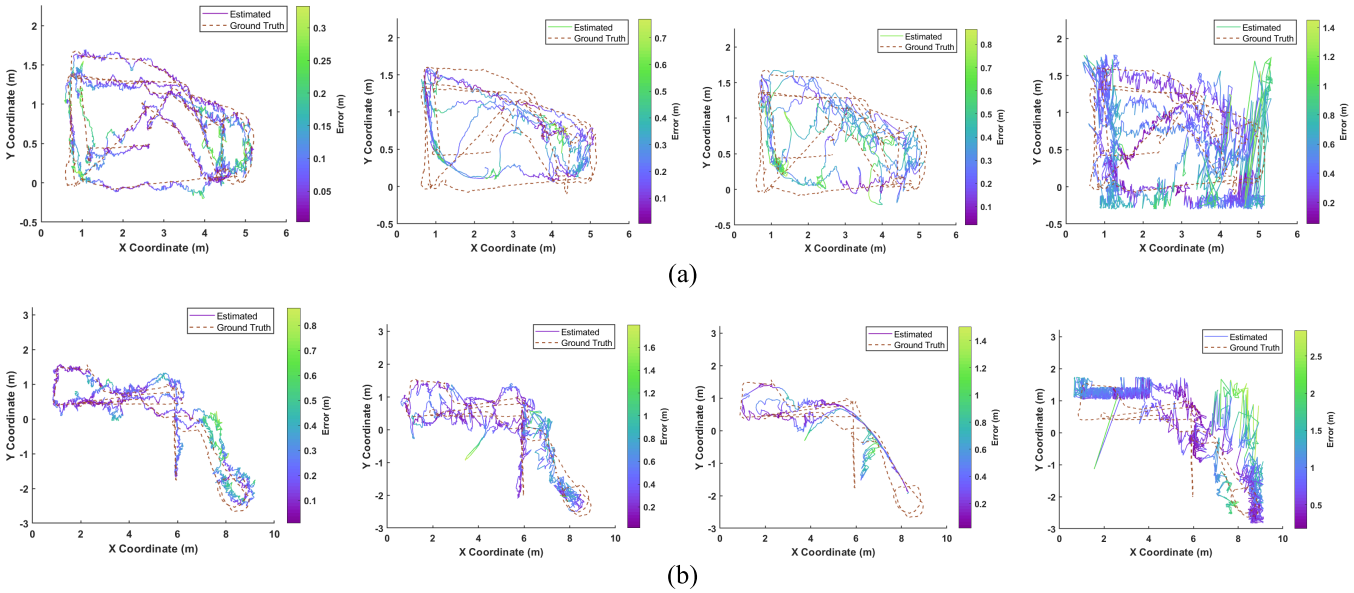
The Table III and Figs. 8 and 9 illustrate clearly that our proposed method achieves the best localization results among all the methods. We attribute this improvement in two aspects: 1) the proposed graph-based strategy and 2) the proposed multiple-tag strategy.

1) Advantages of the Proposed Graph-Based Strategy: First, the Table III and Figs. 8 and 9 show that the traditional GA-based method cannot achieve desirable results, especially in NLOS situations since it is prone to ranging errors. Then, for the deep learning-based method, STA-GNN-S with only $\boldsymbol{L}_p$ loss beats the CNN-LSTM method significantly, especially in the NLOS situation. As an example, the estimated path of CNN-LSTM method in Fig. 8(b) has a large error in the bottom-right corner, and on the contrary, the STA-GNN-S method obtains a better performance shown in Fig. 8(a). This improvement indicates the advantages of the built graph and the proposed STA-GNN-M. Although the CNN and LSTM neural network can also extract the spatial and temporal features, they lack of the use of geometric information which is important for the UWB localization. In contrast, our proposed method builds the spatial–temporal graph according to the topological structure of UWB localization system, thus comprehensively introducing the geometric information into the neural network. Besides, the carefully designed STA-GNN-M not only extracts the high-level spatial–temporal features according to the constructed graph but also suppresses the

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TABLE II
PARAMETERS OF EACH LAYER IN THE STA-GNN-M

Type	Layer	Details	Input feature dimension	Output feature dimension	Output size
Input	Input Layer	Sample Size: 10 Anchor Size: 3 M=4, N=4, T=12	-	-	Samples: (10,12,16) Anchor: (3,12,16)
Achor Embedding	Fully Connected Layer for Sample	-	10	128	(128,12,16)
	Fully Connected Layer for Anchor	-	3	128	(128,12,16)
	Add two features and resize	-	128	128	(12,128,16)
STA Block1	Temporal Gated Convolutional Layer 1	Kernel Size: (3,1) Stride:1	128	128	(10,128,16)
	Gated Linear Unit 1	-	128	64	(10,64,16)
	Spatial Graph Convolutional Layer	Activation: ReLu	64	16	(10,16,16)
	Temporal Gated Convolutional Layer 2	Kernel Size: (3,1) Stride: 1 Activation: ReLu Dropout: 0.1	16	128	(8,128,16)
	Gated Linear Unit 2	-	128	64	(8,64,16)
	Self-Attention Layer	-	64	64	(8,64,16)
STA Block2	Temporal Gated Convolutional Layer 1	Kernel Size: (3,1) Stride: 1	64	128	(6,128,16)
	Gated Linear Unit 1	-	128	64	(6,64,16)
	Spatial Graph Convolutional Layer	Activation: ReLu	64	16	(6,16,16)
	Temporal Gated Convolutional Layer 2	Kernel Size: (3,1) Stride：1 Activation: ReLu Dropout: 0.1	16	128	(4,128,16)
	Gated Linear Unit 2	-	128	64	(4,64,16)
	Self-Attention Layer	-	64	64	(4,64,16)
Regression Layer	Temporal Gated Convolutional Layer 3	Kernel Size: (4,1) Stride: 1 Dropout: 0.1 Activation: ReLu	64	128	(1,128,16)
	Resize	-	-	-	(512,4)
	Fully Connected Layer 1 for Position	Activation: ReLu	512	64	(64,4)
	Fully Connected Layer 2 for Position	-	64	3	(3,4)
	Fully Connected Layer for Distance	-	512	1	(1,4)



(a)

(b)

Fig. 8. Trajectory error curves of our proposed method and competing methods of (a) L-7 sequence in first experiment and (b) N-9 sequence in the second experiment. Our proposed method, STA-GNN-M with single tag method, CNN-LSTM method, and genetic algorithm-based method (left–right). We only give the 2-D trajectories since the changes on the z-axis are not very large.

ranging errors for UWB localization. Therefore, it can achieve a better performance than the CNN-LSTM method.

2) Advantages of the Proposed Multiple-Tag Strategy: On the one hand, Table III clearly shows that the results of

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TABLE III
RMSE OF TESTING SEQUENCES OF OUR PROPOSED METHOD AND COMPETING METHODS (M)

Sequence	STA-GNN-M	STA-GNN-M with L_p	STA-GNN-S with L_p	CNN-LSTM	GA
L-1	**0.051**	0.058	0.119	0.194	0.245
L-2	**0.047**	0.062	0.137	0.196	0.276
L-3	**0.074**	0.092	0.136	0.213	0.234
L-4	**0.053**	0.071	0.132	0.191	0.276
L-5	**0.059**	0.071	0.135	0.209	0.233
L-6	**0.049**	0.061	0.120	0.211	0.205
L-7	**0.071**	0.093	0.169	0.232	0.280
L-8	**0.051**	0.067	0.122	0.210	0.281
N-1	**0.224**	0.242	0.284	0.374	0.701
N-2	**0.161**	0.181	0.231	0.321	0.637
N-3	**0.070**	0.088	0.121	0.143	0.370
N-4	**0.140**	0.154	0.177	0.219	0.332
N-5	**0.081**	0.104	0.143	0.190	0.497
N-6	**0.117**	0.138	0.193	0.258	0.663
N-7	**0.149**	0.165	0.264	0.312	0.696
N-8	**0.142**	0.173	0.262	0.329	0.816
N-9	**0.150**	0.166	0.257	0.268	0.504
N-10	**0.122**	0.134	0.224	0.265	0.561
N-11	**0.108**	0.132	0.226	0.310	0.708

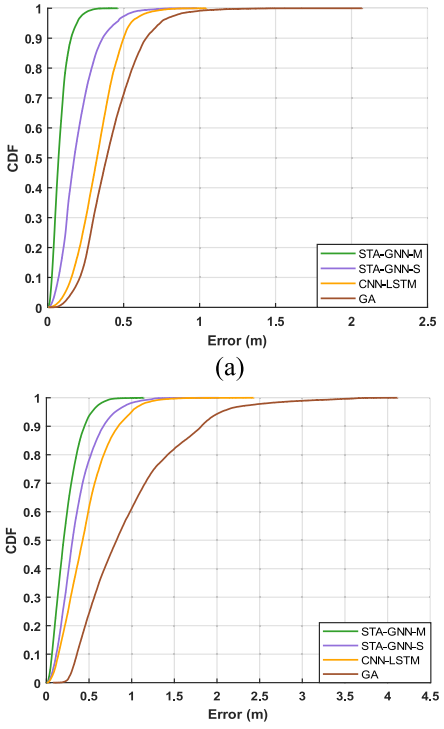


Fig. 9. CDFs of localization errors of (a) whole sequences in first experiment and (b) whole sequences in second experiment.

STA-GNN-M with only L_p loss better than the STA-GNN-S with L_p loss. It improves the accuracy by more than 40% in most sequences. This result suggests that using multiple tags can benefit the UWB localization since the multiple tags provide more ranging measurements. This extra information improves the robustness and the accuracy of the UWB localization.

On the other hand, depending on the geometric information provided by the multiple tags, we also design a novel cost function with four losses to train the network effectively.

In order to show the effect of each loss in the designed cost function, we conducted an ablation experiment to test six different combinations of the losses. The position loss $\boldsymbol{L}_p$ is included in all combinations since our method aims to estimate the tag locations.

The results of this ablation experiment are shown in Table IV. First, the table illustrates that using both the distance loss $\boldsymbol{L}_d$ and position loss $\boldsymbol{L}_p$ improves the performance compared to using only the position loss $\boldsymbol{L}_p$. In addition, the method with cost function, including $\boldsymbol{L}_p$, $\boldsymbol{L}_d$, and $\boldsymbol{L}_m$, also beats the results with only $\boldsymbol{L}_p$ and $\boldsymbol{L}_m$ losses. These results indicate the advantages of distance loss.

Similarly, for multi-tag loss $\boldsymbol{L}_m$, the results of using all four losses are around 7% better on average compared to the cost function without $\boldsymbol{L}_m$, while the performance of using $\boldsymbol{L}_p$, $\boldsymbol{L}_d$, and $\boldsymbol{L}_m$ losses is also better than only using $\boldsymbol{L}_p$ and $\boldsymbol{L}_d$ losses.

Finally, for the estimation loss $\boldsymbol{L}_e$, Table IV shows that using all four losses or using $\boldsymbol{L}_p$, $\boldsymbol{L}_d$, and $\boldsymbol{L}_e$ losses achieves better performance than the method without $\boldsymbol{L}_e$ loss or the method with $\boldsymbol{L}_p$ and $\boldsymbol{L}_d$ losses.

Therefore, for each of the distance loss, the multi-tag loss, and the estimation loss, we both use two comparative experiments to show the advantages of them. This ablation experiment can suggest the effectiveness of the designed cost function.

To sum up, through our experiments, we analyze the advantages of our proposed method, and demonstrate its effectiveness.

C. Performance Analysis of the Network

In this section, we conduct a series of experiments to further analyze the performance of our proposed network under various conditions.

1) Analysis on Network Parameters: In this section, we discuss the performance of our proposed network with different parameter sizes and training dataset sizes. We implement our proposed network with three different parameter sizes: light network, medium network, and large network. Among them, the medium network is the network used in our work with 266 477 parameters. The light network has 68 345 parameters. It only contains one STA block, and the input feature dimension of this network is set as 64. For the large network, the input feature dimension is set as 256, and three STA blocks are included; thus, it has 804 749 parameters. For the training sequences, we take 30%, 60%, 90%, and 100% of the data according to the average distribution manner to train these three types of networks, demonstrating how the size of the training set impacts the performance of models with different complexities in both LOS and NLOS environments.

The average results of all the testing sequences are illustrated in Table V. On the one hand, the performances of all three networks demonstrate an upward trend when the percentage of training sequence size grows from 30% to 100%, suggesting that a sufficient amount of data is necessary to train networks with desirable performance. On the other hand, the light network obtains the worst performance since it cannot capture high-level spatial–temporal features with a limited model scale. In addition, the performance of medium network

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TABLE IV
RMSE OF TESTING SEQUENCES OF OUR PROPOSED METHOD WITH DIFFERENT COST FUNCTIONS (M)

Sequence	Cost Function (with the weights for different loss terms)					
	$\boldsymbol{L_p}$+$\boldsymbol{L_d}$+$\boldsymbol{L_m}$+$\boldsymbol{L_e}$	$\boldsymbol{L_p}$+$\boldsymbol{L_d}$+$\boldsymbol{L_e}$	$\boldsymbol{L_p}$+$\boldsymbol{L_d}$+$\boldsymbol{L_m}$	$\boldsymbol{L_p}$+$\boldsymbol{L_m}$	$\boldsymbol{L_p}$+$\boldsymbol{L_d}$	$\boldsymbol{L_p}$
L-1	0.051	0.053	0.054	0.056	0.055	0.058
L-2	0.047	0.055	0.053	0.058	0.057	0.062
L-3	0.074	0.079	0.082	0.084	0.085	0.092
L-4	0.053	0.060	0.059	0.066	0.062	0.071
L-5	0.059	0.069	0.062	0.069	0.066	0.071
L-6	0.049	0.058	0.054	0.059	0.058	0.061
L-7	0.071	0.078	0.077	0.085	0.084	0.093
L-8	0.051	0.058	0.052	0.065	0.060	0.067
N-1	0.224	0.230	0.232	0.241	0.235	0.242
N-2	0.161	0.165	0.171	0.177	0.179	0.181
N-3	0.070	0.073	0.080	0.073	0.076	0.088
N-4	0.140	0.146	0.140	0.159	0.142	0.154
N-5	0.081	0.083	0.091	0.091	0.099	0.104
N-6	0.117	0.112	0.115	0.119	0.124	0.138
N-7	0.149	0.153	0.168	0.149	0.164	0.165
N-8	0.142	0.150	0.161	0.156	0.169	0.173
N-9	0.150	0.156	0.160	0.165	0.162	0.166
N-10	0.122	0.127	0.129	0.136	0.134	0.134
N-11	0.108	0.110	0.118	0.118	0.123	0.132

TABLE V
COMPARISON BETWEEN THE TRAINING SET SIZE AND TRAINABLE PARAMETERS (M)

Scene	Network Scale	Number of Parameters	Percentage of training sequence size			
			30%	60%	90%	100%
LOS	Light	68345	0.091	0.079	0.074	0.076
	Medium (Used)	266477	0.075	0.061	0.059	**0.056**
	Large	804749	0.083	0.068	0.062	0.061
NLOS	Light	68345	0.187	0.158	0.151	0.149
	Medium (Used)	266477	0.169	0.141	0.135	**0.133**
	Large	804749	0.172	0.146	0.139	0.141

TABLE VI
RMSE OF STA-GNN-M WITH VARYING WEIGHTS FOR L_m AND L_e AND FIXED WEIGHTS FOR $L_p = 1$ AND $L_d = 0.1$

Environment	Weight	α_3=0.1	α_3=0.2	α_3=0.35	α_3=0.5
LOS	α_4 =0.1	0.062	0.060	0.063	0.064
	α_4 =0.2	0.061	0.059	0.064	0.085
	α_4 =0.35	0.064	0.065	0.064	0.096
	α_4 =0.5	0.112	0.103	0.105	0.106
NLOS	α_4 =0.1	0.143	0.141	0.144	0.145
	α_4 =0.2	0.140	0.139	0.142	0.146
	α_4 =0.35	0.152	0.144	0.157	0.178
	α_4 =0.5	0.178	0.181	0.178	0.179

TABLE VII
RMSE OF STA-GNN-M WITH VARYING WEIGHTS FOR L_m AND L_e AND FIXED WEIGHTS FOR $L_p = 1$ AND $L_d = 0.2$

Environment	Weight	α_3=0.1	α_3=0.2	α_3=0.35	α_3=0.5
LOS	α_4 =0.1	0.060	0.059	0.062	0.063
	α_4 =0.2	0.058	**0.056**	0.063	0.060
	α_4 =0.35	0.065	0.067	0.071	0.095
	α_4 =0.5	0.101	0.111	0.084	0.099
NLOS	α_4 =0.1	0.139	0.136	0.141	0.140
	α_4 =0.2	0.137	**0.133**	0.142	0.144
	α_4 =0.35	0.142	0.154	0.157	0.178
	α_4 =0.5	0.183	0.180	0.178	0.181

is slightly better than that of large network. The main reason is that the size of training dataset is insufficient to train the large network. Thus, considering the cost of the training process and the cost of collecting ground-truth values, we use the medium network in this work.

2) Analysis on Weights of Loss Functions: In order to determine the weights of different loss functions, we conduct an experiment with different quadruplets $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ of parameters on both LOS and NLOS sequences. It is clear that $\boldsymbol{L}_p$ should always be included, and thus, we set its weight α_1 as 1. Then, we take different numerical combinations for α_2, α_3, and α_4 to train the whole network, and their average results of all the testing sequences are shown in Tables VI–IX.

Tables VI–IX show that our proposed network reports the comparable performance when α_3 and α_4 are set to less than 0.5, indicating that STA-GNN-M is not significantly affected by weight variations in this situation. In particular, the proposed STA-GNN-M achieves the best performance when $\alpha_2 = 0.2$, $\alpha_3 = 0.2$, and $\alpha_4 = 0.2$. Thus, in this work, we set $\alpha_1, \alpha_2, \alpha_3$, and α_4 as 1, 0.2, 0.2, and 0.2, respectively.

3) Analysis on the Situation When Anchors Are Dropped: Finally, we discuss the performance of our proposed network when an anchor is dropped at the testing data. In this experiment, we consider two situations: 1) the random anchors are dropped: we drop the ranging measurement of a random anchor at a certain percentage of the timesteps according to the average distribution and 2) the fixed anchors are dropped: we drop the ranging measurement of anchor 2 at a certain percentage of the timesteps according to the average distribution. For these dropped values, we use the ranging measurements in the last timestep to pad them. If all the measurements are dropped, the zero is used for padding. The results of these two cases are given in Table X.

Table X shows that whether in the first or second case, the performance of our proposed network is not significantly

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TABLE VIII

RMSE OF STA-GNN-M WITH VARYING WEIGHTS FOR L_m AND L_e AND FIXED WEIGHTS FOR $L_p = 1$ AND $L_d = 0.5$

Environment	Weight	α_3=0.1	α_3=0.2	α_3=0.35	α_3=0.5
LOS	α_4 =0.1	0.066	0.061	0.063	0.064
	α_4 =0.2	0.063	0.063	0.067	0.066
	α_4 =0.35	0.072	0.074	0.076	0.097
	α_4 =0.5	0.102	0.104	0.106	0.113
NLOS	α_4 =0.1	0.143	0.141	0.143	0.144
	α_4 =0.2	0.142	0.140	0.147	0.143
	α_4 =0.35	0.152	0.154	0.158	0.178
	α_4 =0.5	0.169	0.167	0.174	0.175

TABLE IX

RMSE OF STA-GNN-M WITH VARYING WEIGHTS FOR L_m AND L_e AND FIXED WEIGHTS FOR $L_p = 1$ AND $L_d = 1$

Environment	Weight	α_3=0.1	α_3=0.2	α_3=0.35	α_3=0.5
LOS	α_4 =0.1	0.068	0.066	0.067	0.067
	α_4 =0.2	0.067	0.065	0.066	0.065
	α_4 =0.35	0.075	0.077	0.085	0.098
	α_4 =0.5	0.112	0.103	0.105	0.106
NLOS	α_4 =0.1	0.146	0.145	0.148	0.152
	α_4 =0.2	0.143	0.143	0.147	0.145
	α_4 =0.35	0.152	0.154	0.155	0.178
	α_4 =0.5	0.167	0.166	0.173	0.164

TABLE X

RMSE OF STA-GNN-M FOR EACH SEQUENCE UNDER TWO ANCHOR DROPPING METHODS

`Sequence	Without Dropping	Random Dropping		Fixed Dropping		
		100%	90%	100%	90%	80%
L-1	0.051	0.061	0.053	0.113	0.057	0.054
L-2	0.047	0.065	0.049	0.156	0.052	0.049
L-3	0.074	0.092	0.078	0.196	0.083	0.079
L-4	0.053	0.076	0.057	0.165	0.059	0.056
L-5	0.059	0.076	0.061	0.155	0.065	0.062
L-6	0.049	0.069	0.052	0.178	0.054	0.051
L-7	0.071	0.081	0.074	0.182	0.075	0.073
L-8	0.051	0.071	0.055	0.164	0.057	0.053
N-1	0.224	0.232	0.226	0.335	0.229	0.226
N-2	0.161	0.169	0.163	0.280	0.165	0.162
N-3	0.070	0.081	0.073	0.105	0.074	0.072
N-4	0.140	0.150	0.142	0.240	0.149	0.143
N-5	0.081	0.089	0.082	0.112	0.087	0.084
N-6	0.117	0.120	0.119	0.223	0.125	0.120
N-7	0.149	0.156	0.152	0.208	0.156	0.154
N-8	0.142	0.154	0.144	0.194	0.151	0.145
N-9	0.150	0.156	0.152	0.267	0.154	0.153
N-10	0.122	0.131	0.126	0.282	0.129	0.125
N-11	0.108	0.120	0.110	0.209	0.117	0.112

affected in the situation that dropping the ranging measurements from one anchor below 90% of the timestep. Only dropping the ranging measurements from one anchor at all timesteps will have a significant negative impact on the localization results. However, this situation will not happen frequently. Therefore, this experiment further demonstrates the effectiveness and robustness of our proposed method.

V. CONCLUSION

In this article, we propose a multi-tag UWB localization method with STA-GNN-M. For this method, we mainly present two novel strategies.

1) We build a graph structure to express the ranging measurements and carefully design an STA-GNN-M to extract the spatial–temporal features as well as restrain the ranging noise.
2) We use multiple tags in our method and design a novel cost function, which comprehensively uses the geometric information.

The effectiveness of these two strategies is verified through the constructed real-world experiments. The experiments demonstrate that based on these two strategies, the geometric information that is not used sufficiently in the existing method can be fully considered, and thus, the accuracy and robustness of UWB localization in both LOS and NLOS environments are improved.

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